



Executing Party REST API Configuration Specification Annex I

Document Control History

Date	Description	Changed by	Version
17 th October 2019	Initial specification created	PP	1.0
4 th December 2019	Updated to new branding	PP	2.0
16 th December 2019	Updated for Get Orders functionality	PP	2.1
2 nd January 2020	Added detail for APIR fund identifier	PP	2.2
24 th January 2020	Minor changes to fields: - Order Confirm Transaction Type is now Mandatory - Order Reference removed from Order Accept supported field as ID is used instead to relate order instruction to order accept	PP	2.3
2 nd November 2021	Updated example to properly reflect GET order details	PP	2.4
8 th November 2021	Added support for other proprietary fund ID into orders API	PP	2.5
15 th November 2021	Order Confirm - Updated samples to use correct capitalisation of fields - Updates to description for date and date/time formats	PP	2.6
10 th December 2021	Order Status - Expanded detail around rejection fields Get Updated Orders - Added detail regarding suggested use case for these requests - Added detail for pagination	PP	2.7
6 th January 2022	Updated transactionDetails/tradeDateTime content definitions	AB	2.8
4 th Nov 2022	Added information around handling HTTP responses	AB	2.9
19 th January 2023	Removed 'roundingAmount' from Order Instruction message	PP	3.0

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Document Purpose

The purpose of this document is to define the service that Calastone will provide to the Executing Party with respect to managing orders. The document addresses how Calastone will handle mutual fund orders.

Although this document captures the initial requirements for the Executing Party, this is a working document that will be updated as the Ordering Party or Executing Party expands the scope of its use of Calastone. Each future request for change will need to be agreed with Calastone so that all parties explicitly document new requirements and how they can be tested prior to the Production service change.

Disclaimer

The contents of each message outlined in the following section are the extent of fields supplied by current Ordering Parties and Executing Parties. Not all fields are mandatory and therefore not supported by every Executing Party. If a field is provided by an Ordering Party that an Executing Party is not able to receive, the order and any updates relating to it will be passed on without the unsupported fields. It is the responsibility of the Ordering Party and Executing Party to agree which fields will be supported before going live.

Any additional fields supplied that do not form part of this specification will be ignored by Calastone and will not be sent on to the counterparty.

REST API Overview

The Calastone REST API enables the Executing Party to connect to the Calastone system to utilise different products and services.

The API works on a request/response basis using the HTTP protocol and data exchange will be in JSON format.

In order to connect to the Calastone REST API, the client must obtain a token from Calastone. This will be done sending a Token Request to obtain a token which must then be supplied within the HTTP header of any subsequent messages. The token must be refreshed on a periodic basis.

Request Types

The request types supported by the REST API are outlined in the below table.

Type	Description
Get Token	Requests a token from Calastone.
Get Orders	Retrieves a list of orders which have been created within the parameters supplied in the request.
Order Status	Allows the accept or reject of an order request, including supplying additional information
Order Confirm	Allows the confirmation of an order request, providing pricing/charges and realization of the order request

It is expected that the scope of this list may be expanded over time in line with the business requirements of the Executing Party.

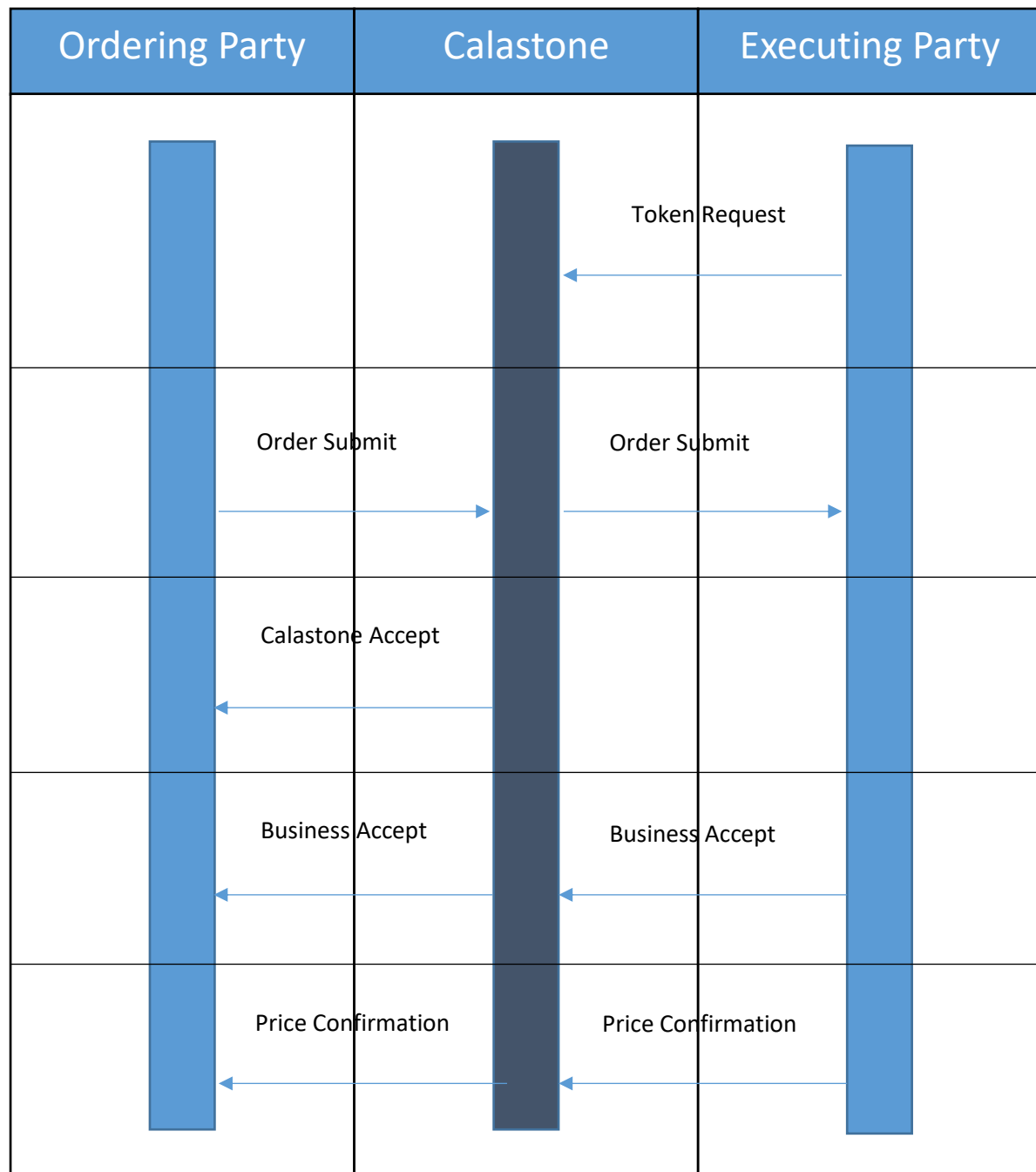
Order Instruction and Update Types

The order instruction and update types supported by the REST API are outlined in the below table.

Type	Description
Subscription Order	Instructs the Executing Party to subscribe to one or more financial instruments, for the same account.
Redemption Order	Instructs the Executing Party to redeem from one or more financial instruments, for the same account.
Switch Order	Instructs the Executing Party to switch: - A specified amount or quantity <i>from</i> specified financial instrument(s) - A specified percentage of the redemption leg <i>to</i> financial instrument(s). These must total 100%.
Get Orders Report	This shows all orders requests submitted since a particular time
Order Instruction Status Report	This gives the status of an order instruction that was previously sent.

Operating Model

The following diagram represents the operational flow for the instruction and execution of an order.



Token Request

Access to the Orders API requires each request to include a bearer token (JWT) from the Calastone Identity Provider Service. The “client” and “client_secret” values will be provided by Calastone.

Token Request Example

The below table provides an example of a token request and the synchronous response provided by the API.

Request	POST / https://login.calastone.com/oauth/connect/token
Body	grant_type=client_credentials&scope=orderapi&client_id= client &client_secret= secret
Response	200, OK
Body	{ "access_token":"eyJhbGciOiJIUyI6IjE1NTQzNz.itgCx_2uzrfez9ue", "expires_in":3600, "token_type":"Bearer" }

The bearer token must be included in the “Authorisation” header for all requests to the Orders API.

Get Orders

To show instructed orders from an Ordering Party, a two-stage process using the Get Orders type request process must be made using the token from the Token Request.

Firstly, by downloading a list of orders that have been created. There are two options for this:

- The relevant time parameter should be contained within the request, which will vary depending on the polling frequency being undertaken. Note that the date/time has a maximum range of 5 days
- No time parameter is specified – this will retrieve a list of all orders updated to their current state in the last 5 days

Secondly, where the order details are to be downloaded for processing, should be retrieved using a Get Orders request where the Order ID is specified.

Each executing party has the responsibility to ensure that when retrieving orders from the API that these are only processed once. Calastone will not have visibility on what orders have been retrieved and processed by executing parties so therefore will not exclude them from further Get Updated Orders Requests of either type.

The maximum number of order records that can be displayed for the results of each query is 500. Additional records to be retrieved can be achieved using pagination. The process of how to use pagination within Orders API will be detailed in this section.

Suggested Use Cases

When the executing party wants to regularly check for new order instructions being sent to them, frequent polling should be undertaken. The suggested approach for this is to use the Get Updated Orders Request with a Time Parameter set.

For example, if polling every 5 minutes for the latest order instructions, the date time could be set for 5 minutes plus 1 minute in the past.

This will only bring back the latest order instructions and reduce the overhead on the executing party's system.

If an executing party wants to download a list of trades for oversight or control purposes, either use the Get Updated Request with a Time Parameter set back to the point of review. Examples:

- if a list of all trades received in the last 24 hours is needed, set the date time parameter to 1 working day earlier with the current time value
- if a list of all trades received in the current business day is needed, set the date time parameter to the current date and the time to "000000Z"

Or the executing party can use the Get Updated Orders query without specifying a date/time parameter. Note that this will bring back the last 5 calendar days' worth of trades, which could be a significant volume depending on the business flows.

It is not recommended that an executing party use the Get Updated Orders query without a date/time parameter for regular polling of the orders API. Due to the 5 day range and the potential for high volumes this could result in a large data set being returned, which will add overhead to the executing party's system.

Pagination

When using either Get Updated Orders queries, the maximum number of records returned in the body of the query is 500 orders.

In order to support queries where the number of returned order entries that exceeds 500, the Get Updated Orders query has pagination functionality.

To identify if pagination is in effect, the header of the returned JSON message will include details about paging for the query. This is the “X-Pagination” header.

The links for previous/next page are only provided where relevant, otherwise 'null' is returned.

Example of the “X-Pagination Header” where the number of records exceeds 500:

```
{
  "totalCount":600,
  "pageSize":500,
  "currentPage":1,
  "totalPages":2,
  "previousPageLink":null,
  "nextPageLink":"https://<BASEURL>/api/orders?updated=20190517T010000Z&pageNumber=2&pageSize=500"
}
```

Example of the “X-Pagination” header where the number of records does not exceed 500:

```
{
  "totalCount":150,
  "pageSize":500,
  "currentPage":1,
  "totalPages":1,
  "previousPageLink":null,
  "nextPageLink":null
}
```

To access the remaining pages the “nextPageLink” URL should be used until the “currentPage” value equals the “totalPages” value. Also, at the last page, the “nextPageLink” will be ‘null’.

Get Updated Orders Request Example (Time Parameter Set)

The below table provides an example of a Get Updated Orders request within a date/time range and the synchronous response provided by the API.

Request	GET /orders?updated={yyyyMMddHHmmssZ}
Body	None
Response	200, OK
Body	<pre>[{ "id": "9ad81cba-2a87-4863-87b7-61600f736cd5", "orderReference": "TestRed019", "status": "Confirmed", "createdUtc": "2019-05-01T15:42:53.6322757+00:00", "updatedUtc": "2019-05-01T15:45:07.92412+00:00" }, { "id": "b76d247d-f7a2-4d2e-8dcd-50b1ab208e40", "orderReference": "TestSw010", "status": "Confirmed", "createdUtc": "2019-05-01T15:30:20.7824982+00:00", "updatedUtc": "2019-05-01T15:32:34.704384+00:00" }, { "id": "9e1eb644-2562-49d6-a864-5bd17d82649b", "orderReference": "TestRed018", "status": "Invalid", "createdUtc": "2019-05-01T15:13:31.2474561+00:00", "updatedUtc": "2019-05-01T15:13:43.012758+00:00" }]</pre>

Get Updated Orders Request Example (No Time Parameter Set)

The below table provides an example of a Get Updated Orders request and the synchronous response provided by the API for all orders updated in the last 5 days.

Request	GET /orders
Body	None
Response	200, OK
Body	<pre>[{ "id": "9ad81cba-2a87-4863-87b7-61600f736cd5", "orderReference": "TestRed019", "status": "Confirmed", "createdUtc": "2019-05-01T15:42:53.6322757+00:00", "updatedUtc": "2019-05-01T15:45:07.92412+00:00" }, { "id": "b76d247d-f7a2-4d2e-8dcd-50b1ab208e40", "orderReference": "TestSw010", "status": "Confirmed", "createdUtc": "2019-05-01T15:30:20.7824982+00:00", "updatedUtc": "2019-05-01T15:32:34.704384+00:00" },]</pre>

There are various status that will be shown against orders in the last updated:

Status	Description
Submitted	Order instruction has been received, passed Calastone validation and is ready to be picked up for processing
Accepted	Order has been accepted by the Executing Party (instruction already processed)
Rejected	Order has been rejected by the Executing Party (instruction already processed)
Confirmed	An accepted order has been updated with completed trade information (fully processed state)

For the purposes of processing new instruction requests the Executing Party should be looking for items in a **Submitted** state only.

The below table provides an example of a retrieving the details of order instructions where the order reference is specified, with the synchronous response provided by the API.

Request	GET /orders/{id}
Body	<i>n/a</i>
Response	200, OK
Body	<pre>[{ "id": "2bf8f2d1-de17-401f-9e7a-0ab4194567c1", "status": "Submitted", "createdUtc": "2019-10-08T12:53:58.5303505+00:00", "updatedUtc": "2019-10-08T12:56:56.2483588+00:00", "submitDetails": { "orderType": "SubscriptionOrder", "orderReference": "JbRedTest166", "fundProviderIdentifier": "STANCHART", "createdUtc": "2019-10-08T12:53:58.5303505+00:00", "transactions": [{ "transactionType": "Subscription", "fundIdentifier": { "type": "ISIN", "identifier": "GB0001234567", "name": "Example", "supplementaryIdentifier": "Code I" }, "orderAmount": { "amount": 1000.0, "currency": "GBP" }, "requestedSettlementCurrency": "GBP", "requestedNavCurrency": "GBP", "accountDetails": {</pre>

	<pre> "accountId": "Account1", "subAccount": "Account2", "agentCode": "Account3" }, "roundingDirection": "Up" }] }, "validationDetails": { "valid": true, "createdUtc": "2019-10-08T12:56:56.2483588+00:00", "orderReference": "JbRedTest166" } }] </pre>
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The below table provides an example of a retrieving the details of order instructions where the ID is specified, with the synchronous response provided by the API.

Request	GET /orders/{id}
Body	<i>n/a</i>
Response	200, OK
Body	<pre> { "id": "2bf8f2d1-de17-401f-9e7a-0ab4194567c1", "status": "Submitted", "createdUtc": "2019-10-08T12:53:58.5303505+00:00", "updatedUtc": "2019-10-08T12:56:56.2483588+00:00", "submitDetails": { "orderType": "SubscriptionOrder", "orderReference": "JbRedTest166", "fundProviderIdentifier": "STANCHART", "createdUtc": "2019-10-08T12:53:58.5303505+00:00", "transactions": [{ "transactionType": "Subscription", "fundIdentifier": { "type": "ISIN", "identifier": "GB0001234567", "name": "Example", "supplementaryIdentifier": "Code I" }, "orderAmount": { "amount": 1000.0, "currency": "GBP" }, "requestedSettlementCurrency": "GBP", "requestedNavCurrency": "GBP", "accountDetails": { </pre>

	<pre> "accountId": "Account1", "subAccount": "Account2", "agentCode": "Account3" }, "roundingDirection": "Up" }] }, "validationDetails": { "valid": true, "createdUtc": "2019-10-08T12:56:56.2483588+00:00", "orderReference": "JbRedTest166" } } </pre>
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Order Instruction Request

The following table outlines the contents of an Order Instruction request.

Switch	Subscription/ Redemption	JSON Field Name	Business Attribute Name	Business Attribute Definition	Field Format
M	M	orderType	Order/Switch Identifier	Identifies whether the order is a subscription, redemption or switch.	Type: Code Set Format: MaxLength; 17 Content: Subscription: "SubscriptionOrder" Redemption: "RedemptionOrder" Switch: "SwitchOrder"
M	M	orderReference	Order Reference Switch Reference	Unique and unambiguous identifier for an order, as assigned by the Ordering Party.	Type: String Format: MaxLength; 35 Content: Unique Order Reference
M	M	fundProviderIdentifier	Fund Provider Identifier ⁶	The Executing Party (e.g. the fund provider) as identified by the Ordering Party.	Type: Code Set Format: MaxLength; 35 Content: Pre-defined identifier as agreed between Calastone and Ordering Party
M	-	transactions/legId	Switch: Leg Order Reference	Unique and unambiguous identifier for the subscription/redemption leg of a switch order, as assigned by the Ordering Party.	Type: String Format: MaxLength; 35 Content: Unique Order Reference
M	M	transactions/transactionType	OrderType	Identifies whether the order type of the switch leg or the order type.	Type: Code Set Format: MaxLength; 12 Content: Subscription/Subscription Leg: "Subscription" Redemption/Redemption Leg: "Redemption"
M	M	transactions/fundIdentifier/type	Fund ID Type ^{4,9}	Identifies the type of identifier used in FndId Column	Type: Code Set Format: MaxLength; 5 Content: 'ISIN', 'SEDOL', 'APIR', 'CUSIP', 'OTHER'

C ¹⁰	C ¹⁰	Transactions/fundIdentifier/ otherType	Proprietary Fund ID Source ⁹	Identifies the designation of fund identifier type when Fund ID Type = "OTHER" Used for fund identifiers outside of ISIN, SEDOL, APIR and CUSIP	Type: String Format: MaxLength; 35 Content: Pre-defined fund identifier type agreed between the Ordering Party and the Executing Party, outside of supported values under "Fund ID Type"
M	M	transactions/fundIdentifier/ identifier	Fund ID ⁴	Identifier of the fund. Type of identifier used is defined in FundIDType column	Type: String Format: For ISIN; exactly 12 characters. For SEDOL; exactly 7 characters. For APIR; exactly 9 characters For CUSIP; exactly 9 characters For OTHER; up to 35 characters Content: Valid fund identifier
O	O	transactions/fundIdentifier name	Fund Name ³	The name of the fund for which the order is being placed.	Type: String Format: MaxLength; 105 Content: Fund Name
O	O	transactions/fundIdentifier/ supplementaryIdentifier	Product Code ³	Additional information about a financial instrument to help identify the instrument or the code of the product, which is a subdivision of a fund.	Type: String Format: MaxLength; 28 Content: Product Code
{M or	{M or	transactions/unit	Units ¹	Quantity expressed as a number of units.	Type: Decimal Number Format: MaxFractionDigits; 17 MaxTotalDigits; 18 Content: Unit Amount
M or	M	transactions/orderAmount/ amount	Ordered Amount ^{1,2}	Quantity expressed as a monetary amount	Type: Decimal Number Format: MaxFractionDigits; 17 MaxTotalDigits; 18 Content: Order Amount
	M	transactions/orderAmount/ currency	Ordered Currency ^{1,2}	The currency indicator for the ordered amount.	Type: String Format: Length; 3 Content: ISO Currency Code
Or M}	Or M}	transactions/ holdingsRedemptionRate	Percentage of Holdings to be Redeemed ¹	Indicates the percentage to be redeemed. Only 100% switches and redemptions are currently supported.	Type: Integer Format: Length; 3 Content: '100'
M	-	transactions/ percentageOfRedemptionAmount	Percentage of Redemption Amount to be Switched	Indicates the percentage to be redeemed. Only 100% switches and redemptions are currently supported.	Type: Integer Format: Length; 3 Content: '100' in the subscription leg only.

O		O		transactions/ requestedSettlementCurrency	Requested Settlement Currency ³	The requested settlement currency for the transaction.	Type: String Format: Length; 3 Content: ISO Currency Code
O		O		transactions/ requestedNavCurrency	Requested Pricing Currency ³	The preferred currency for the pricing of the fund. This must be one of the available currencies, as defined in the fund's prospectus.	Type: String Format: Length; 3 Content: ISO Currency Code
O	M	O	M	transactions/chargeDetails/ type	Charge Type ³ Charge Extended Type ³	Service for which a charge is asked or paid, eg, brokerage fees.	Type: Code Set Format: Length; 4 Content: ISO 20022 Code
	M		M	transactions/chargeDetails/ rate	Charge Rate ³	Rate used to calculate the amount of the charge or fee expressed as a percentage.	Type: Decimal Number Format: MaxFractionDigits; 10 MaxTotalDigits: 11 Content: Charge Rate as percentage
O	M	O	M	transactions/commissionDetails/typ e	Commission Type ³ Commission Extended Type ³	Service for which the commission/fee is asked or paid	Type: Code Set Format: Length; 4 Content: ISO 20022 Code
	M		M	transactions/commissionDetails/rat e	Commission Rate ^{3,8}	Commission/Fee expressed as a percentage.	Type: Decimal Number Format: FractionDigits; 10 MaxDigits: 11 Content: Commission rate as Percentage
				transactions/commissionDetails/ percentageWaived	Percentage Commission Waived ^{3,8}	Proportion of the commission that is waived as a percentage.	Type: Decimal Number Format: FractionDigits; 10 MaxDigits: 11 Content: Commission Waived proportion as percentage
M		M		transactions/accountDetails/ accountId	Account ⁷	The unique and unambiguous identification for the account between the Ordering Party and the Executing Party. This is normally issued by the Executing Party.	Type: String Format: MaxLength; 35 Content: Account Id
O		O		transactions/accountDetails subAccount	Sub Account ^{3,5,7}	Subdivision of an account used to segregate specific holdings.	Type: String Format: MaxLength; 35 Content: Sub Account Id
O		O		transactions/accountDetails/ agentCode	Agent Code ^{5,7}	The identification for the Ordering Party as issued by the Executing Party.	Type: String Format: MaxLength; 35 Content: Agent code

○	○	transactions/rounding Direction	Rounding Direction	Rounding direction applied to fractional numbers.	Type: Code Set Format: MaxLength; 4 Content: Round up: “Up”, round down: “Down”
○	○	group1Or2Units	Group 1 or 2 Identifier ³	Tax group to which the purchased investment fund units belong. This can be either group 1 or 2.	Type: Code Set Format: MaxLength; 6 Content: “Group1”, “Group2”

¹ For subscriptions, either Units, or Ordered Amount and Ordered Currency, must be supplied in the order. For redemptions, either Units, or Ordered Amount and Ordered Currency, or Percentage of Holdings to be Redeemed, must be supplied in the order. If Percentage of Holdings to be Redeemed is supplied, only 100% redemptions are currently supported.

² Based on previous experience, this Business Attribute can be interpreted differently by counterparties. You should verify your interpretation of it with each Ordering Party before connecting to them.

³ This Business Attribute is not supported by all counterparties. You should agree whether this Business Attribute can be supported by each Ordering Party before connecting to them and agreeing to receive instructions electronically. If an order is received from an Ordering Party with this Business Attribute, CTN will send it on to the Executing Party. If the Executing Party is unable to receive this Business Attribute, CTN will send on the order without the Business Attribute they are unable to receive.

⁴ ISIN, SEDOL, APIR or CUSIP must be used to identify the fund. Not all counterparties support all available fund identifiers. You should agree whether either ISIN, SEDOL, APIR or CUSIP can be supported by each Ordering Party before connecting to them. If an order received from an Ordering Party contains either an ISIN, SEDOL, APIR or CUSIP, CTN will send it on to the Executing Party.

⁵ This Business Attribute is mandatory where required by the Executing Party. You should agree whether it must be supplied with each Executing Party before connecting to them.

⁶ Calastone does not hold any static data, therefore an identifier is required in the order to route the message to the correct Fund Provider. The Fund Provider Identifier can either be one of the Calastone standard identifiers or one of the Ordering Party's identifiers. These must be agreed in advance with Calastone before going live with each Executing Party.

⁷ For switches, Account, Agent Code and Sub Account details (where required) must be contained on both legs of the order.

⁸ Either Commission Rate, or Commission Waived Rate, but not both, may be supplied in the order.

⁹ Use of OTHER, specifically the Proprietary Fund ID Source used, may not be supported by all counterparties. You agree should whether the type of fund identifier (outside of ISIN, SEDOL, APIR or CUSIP) can be supported by each Ordering Party before connecting to the them. If an order is received from an Order Party containing a fund ID designated under OTHER, CTN will send it on to the Executing Party

¹⁰ Must be supplied if field 'transactions/fundIdentifier/type' = "OTHER". Else must not be supplied

Order Status Update

This section covers how to respond to an order instruction request via the API either by:

- Accepting the order request and providing an deal reference
- Rejecting the order request and providing a rejection reason

In both instances, the ID must be specified in the request when accepting or rejecting an instruction.

For an accept update, the following data can be provided:

Business Attribute Name	Business Attribute Definition	Mandatory/Optional	Field Format
dealReference	Trade reference supplied by the executing party	M	Type: String Format: MaxLength; 350 Content: Reference as free text
expectedTradeDateTime	Trade date for order	O	Type: Date Format: Length; 10 Content: YYYY-MM-DD
expectedCashSettlementDate	Expected settlement date for order	O	Type: Date Time Format: Length; 10 Content: YYYY-MM-DD

For a rejection update, the following data must be provided:

Business Attribute Name	Business Attribute Definition	Mandatory/Optional	Field Format
rejectReason	Rejection Reason	M	Type: String Format: MaxLength; 350 Content: Reason as free text

Order Status Examples

The below table provides an example of an executing party accepting an order instruction by the API.

Request	POST /orders/{id}/accept
Body	<pre>[{ "dealReference": "DealRef0001", "expectedTradeDateTime": "2019-11-01", "expectedCashSettlementDate": "2019-11-01", }]</pre>
Response	200, OK
Body	<pre>[{ "id": "5925c509-50e2-40f4-ae17-dbb6ce213dac", "status": "Accepted", ... }]</pre>

The below table provides an example of an executing party rejecting an order instruction by the API.

Request	POST /orders/{id}/reject
Body	[{ "rejectReason": "Unknown ISIN" }]
Response	200, OK
Body	[{ "id": "5925c506-50e2-40f4-ae17-dbb6ce213ddc", "status": "Rejected", }]

Order Confirm

This is to provide the executing party the ability to pass priced/confirmed order details back to the ordering party.

The relevant order identifier should be contained within the request in order to access the details of the relevant order instruction.

Order Confirmation Details

Switch		Subscription/ Redemption		JSON Field Name	Business Attribute Name	Business Attribute Definition	Notes/Content
M		M		orderReference	Order Reference Switch Reference	Unique and unambiguous identifier for an order, as assigned by the Ordering Party.	Type: String Format: MaxLength; 35 Content: Unique Order Reference
O		O		transactionDetails\ legId	Switch: Leg Order Reference	Unique and unambiguous identifier for the subscription or redemption leg of a switch order, as assigned by the Ordering Party.	Type: String Format: MaxLength; 35 Content: Leg ID
M		M		transactionDetails\ trasactionType	TransactionType	Identifies whether the order is a subscription, redemption or switch.	Type: Code Set Format: MaxLength; 12 Content: "Subscription" or "Redemption"
M		M		transactionDetails\ dealReference	Sub/Red: Deal Reference Switch: Leg Deal Reference	The unique reference for the transaction assigned by the Executing Party	Type: String Format: MaxLength; 35 Content: Unique Deal Reference
M		M		transactionDetails\ units	Units	Quantity expressed as a number of units.	Type: Decimal Number Format: MaxFractionDigits; 17 MaxTotalDigits: 18 Content: Unit Amount
O		O		transactionDetails\ cashSettlementDate	Settlement Date	The date on which cash is available for settlement.	Type: Date price Format: Length; 10 Content: yyyy-MM-dd
O	M	O	M	transactionDetails\ net Amount\	Net Amount ^{1,2}	Net amount as supplied by Executing Party	Type: Decimal Number Format: MaxFractionDigits; 5

				amount			MaxTotalDigits; 18 Content: Net Amount
	M		M	transactionDetails\netAmount\ncurrency	Net Amount Currency ^{1,2}	The currency that the net amount is provided in.	Type: String Format: Length; 3 Content: ISO Currency Code
O	M	O	M	transactionDetails\grossAmount\amount	Gross Amount ^{1,2}	Gross amount as supplied by Executing Party	Type: Decimal Number Format: MaxFractionDigits; 5 MaxTotalDigits; 18 Content: Gross Amount
	M		M	transactionDetails\grossAmount\ncurrency	Gross Amount Currency ^{1,2}	The currency that the gross amount is provided in.	Type: String Format: Length; 3 Content: ISO Currency Code
-		M		transactionDetails\settlementAmount\amount	Settlement Amount ²	Total amount of money paid /to be paid or received in exchange for the financial instrument in the individual order. The settlement amount as provided by the Executing Party.	Type: Decimal Number Format: MaxFractionDigits; 13 MaxTotalDigits; 18 Content: Settlement Amount
-		M		transactionDetails\settlementAmount\ncurrency	Settlement Amount Currency ²	The settlement currency for the transaction	Type: String Format: Length; 3 Content: ISO Currency Code
M		M		transactionDetails\fundIdentifier\type	Fund ID Type ^{4,9}	Identifies the type of identifier used in FndId Column	Type: Code Set Format: MaxLength; 5 Content: 'ISIN', 'SEDOL', 'APIR', 'CUSIP', 'OTHER'
C ¹⁰		C ¹⁰		Transactions/fundidentifier/otherType	Proprietary Fund ID Source ⁹	Identifies the designation of fund identifier type when Fund ID Type = "OTHER" Used for fund identifiers outside of ISIN, SEDOL, APIR and CUSIP	Type: String Format: MaxLength; 35 Content: Pre-defined fund identifier type agreed between the Ordering Party and the Executing Party, outside of supported values under "Fund ID Type"

M		M		transactionDetails\fundIdentifier\identifier	Fund ID ⁴	Identifier of the fund. Type of identifier used is defined in FndIdType column	Type: String Format: For ISIN; exactly 12 characters For SEDOL; exactly 7 characters For APIR; exactly 9 characters For CUSIP; exactly 9 characters For OTHER; up to 35 characters Content: Valid fund identifier
O		O		transactionDetails\fundIdentifier\name	Fund Name ³	The name of the fund for which the order is being placed.	Type: String Format: MaxLength; 105 Content: Fund Name
O		O		transactionsDetails\fundIdentifier\supplementaryIdentifier	Product Code ³	Additional information about a financial instrument to help identify the instrument or the code of the product, which is a subdivision of a fund.	Type: String Format: MaxLength; 28 Content: Product Code
M		M		transactionDetails\dealingPriceDetails\type	Price Type ^{2,3,6}	The type and information about the price (e.g. Bid, Offer, NAV).	Type: Code Set Format: MaxLength; 355 Content: ISO20022 Code or 'OTHR- <i>{free text}</i> ' if unknown
M		M		transactionDetails\dealingPriceDetails\amount	Price Amount	The value of the price applied to the transaction.	Type: Decimal Number Format: MaxFractionDigits; 13 MaxTotalDigits; 18 Content: Price Amount
M		M		transactionDetails\dealingPriceDetails\currency	Price Currency	The currency of the price applied to the transaction.	Type: String Format: Length; 3 Content: ISO Currency Code
O	M	O	M	transactionDetails\chargeDetails\type	Charge Type ^{2,3,5,6} Extended Charge Type ^{2,3,5,6}	Service for which a charge is asked or paid, eg, brokerage fees.	Type: Code Set Format: MaxLength; 355 Content: ISO20022 Code or 'OTHR- <i>{free text}</i> ' if unknown
	O		O	transactionDetails\chargeDetails\rate	Charge Rate ^{2,3,5}	Rate used to calculate the amount of the charge or fee as a percentage.	Type: Decimal Number Format: MaxFractionDigits; 10 MaxTotalDigits: 11 Content: Charge Rate as percentage
	M		M	transactionDetails\chargeDetails\amount	Charge Amount ^{2,3,5}	Amount of money asked or paid for the charge.	Type: Decimal Number Format: FractionDigits; 13 MaxDigits; 18 Content: Charge Amount

	M		M	transactionDetails\chargeDetails\currency	Charge Amount Currency ^{2,3,5}	The currency of the charge amount to be applied to the transaction.	Type: String Format: Length; 3 Content: ISO Currency Code
O	O	O	O	transactionDetails\commissionDetails\type	Commission Type ^{2,3,5,6} Extended Commission Type ^{2,3,5,6}	Service for which a charge is asked or paid, eg, brokerage fees.	Type: Code Set Format: MaxLength; 355 Content: ISO20022 Code or 'OTHR-{free text}' if unknown
	O		O	transactionDetails\commissionDetails\rate	Commission Rate ^{2,3,5}	Rate used to calculate the amount of the charge or fee as a percentage.	Type: Decimal Number Format: MaxFractionDigits; 10 MaxTotalDigits; 11 Content: Charge Rate as percentage
	M		M	transactionDetails\commissionDetails\amount	Commission Amount ^{2,3,5}	Amount of money asked or paid for the charge.	Type: Decimal Number Format: FractionDigits; 13 MaxDigits; 18 Content: Charge Amount
	M		M	transactionDetails\commissionDetails\currency	Commission Amount Currency ^{2,3,5}	The currency of the charge amount to be applied to the transaction.	Type: String Format: Length; 3 Content: ISO Currency Code
O	M	O	M	transactionDetails\taxDetails\type	Tax Type ^{2,3,5,6}	The type of tax to be applied to the transaction.	Type: Code Set Format: MaxLength; 355 Content: ISO20022 Code or 'OTHR-{free text}' if unknown
	O		O	transactionDetails\taxDetails\rate	Tax Rate ^{2,3,5}	Rate used for calculation of the tax.	Type: Decimal Number Format: MaxFractionDigits; 10 MaxTotalDigits; 11 Content: Tax rate as Percentage
	M		M	transactionDetails\taxDetails\amount	Tax Amount ^{2,3,5}	The amount of the tax to be applied to the transaction.	Type: Decimal Number Format: MaxFractionDigits; 13 MaxTotalDigits; 18 Content: Tax Amount
	M		M	transactionDetails\taxDetails\currency	Tax Amount Currency ^{2,3,5}	The currency of the tax amount to be applied to the transaction.	Type: String Format: Length; 3 Content: ISO Currency Code
	O		O	transactionDetails\taxDetails\exemptionIndicator	Tax Exemption Indicator ^{2,3,5}	Indicates whether a tax exemption applies.	Type: Code Set Format: MaxLength; 3 Content: 'YES' or 'NO'

	O		O	transactionDetails\taxDetails\exemptionReason	Tax Exemption Reason ^{2,3,5}	Reason for a tax exemption.	Type: Code Set Format: MaxLength; 355 Content: ISO20022 Code or 'OTHR-{free text}' if unknown
M		M		transactionDetails\accountDetails\accountId	Account	The unique and unambiguous identification for the account between the Ordering Party and the Executing Party. This is normally issued by the Executing Party.	Type: String Format: MaxLength; 35 Content: This is the account as provided by the Executing Party.
O		O		transactionDetails\accountDetails\subAccount	Sub Account ³	Subdivision of an account used to segregate specific holdings.	Type: String Format: MaxLength; 35 Content: Sub Account Id
O		O		transactionDetails\accountDetails\agentCode	Agent Code	The identification for the Ordering Party as issued by the Executing Party.	Type: String Format: MaxLength; 35 Content: Agent code
M		M		transactionDetails\tradeDateTime	Trade Date Time ⁸	The date and time the transaction was executed by the fund provider or TPA and the price applied. This information is provided by the Executing Party. Please note the time aspect of this field is not mandatory.	Type: Date Time Format: Length; 19 Content: yyyy-MM-ddTHH:mm:ss or yyyy-MM-dd
O	O	O	O	fxDetails\toAmount\amount	To Amount ^{2,3,8}	Cash amount resulting from an FX trade.	Type: Decimal Number Format: MaxFractionDigits; 13 MaxTotalDigits; 18 Content: To Amount
	O		O	fxDetails\toAmount\currency	To Amount Currency ^{2,3,7}	Currency that the cash amount resulting from an FX trade is provided in.	Type: String Format: Length; 3 Content: ISO Currency Code
	O		O	fxDetails\fromAmount\amount	From Amount ^{2,3,7}	Cash amount for which an FX trade is required.	Type: Decimal Number Format: MaxFractionDigits; 13 MaxTotalDigits; 18 Content: From Amount

	O		O	fXDetails\fromAmountCurrency	From Amount Currency ^{2,3,7}	Currency that the cash amount for which an FX trade is required is provided in.	Type: String Format: Length; 3 Content: ISO Currency Code
	M		M	fXDetails\unitCurrency	Unit Currency ^{2,3}	Currency in which the rate of exchange is expressed in a currency exchange.	Type: String Format: Length; 3 Content: ISO Currency Code
	M		M	fXDetails\quotedCurrency	Quoted Currency ^{2,3}	Currency into which the base currency is converted. This will be the order currency.	Type: String Format: Length; 3 Content: ISO Currency Code
	M		M	fXDetails\exchangeRate	Exchange Rate ^{2,3}	ExchangeRate expresses the ratio between UnitCurrency and QuotedCurrency (ExchangeRate = UnitCurrency/QuotedCurrency) as a percentage.	Type: Decimal Number Format: MaxFractionDigits; 10 MaxTotalDigits: 11 Content: Exchange Rate as percentage

¹ Either Net Amount (and Net Amount Currency) or Gross Amount (and Gross Amount Currency) or both will be provided.

² Based on previous experience, this Business Attribute can be interpreted differently by counterparties. You should verify your interpretation of it with each Executing Party before connecting to them.

³ This Business Attribute is not supported by all counterparties. You should agree whether this Business Attribute can be supplied by each Executing Party before connecting to them. If a Price Confirmation is received from an Executing Party with this Business Attribute, Calastone will send it on to the Ordering Party.

⁴ ISIN, SEDOL or APIR must be used to identify the fund. Not all counterparties support all available fund identifiers. You should agree whether either ISIN, SEDOL or APIR can be supported by each Ordering Party before connecting to them. If an order received from an Ordering Party contains either an ISIN, SEDOL or APIR, CTN will send it on to the Executing Party.

⁵ This Business Attribute can be supplied multiple times. Before going live with an Executing Party you should check that you are able to support the number of instances of this attribute that they wish to send. Calastone will not be able to pass on additional instances of this attribute that the Ordering Party is not able to receive as part of the Order Confirmation message and Calastone will not communicate this to the Ordering Party.

⁶ If provided by the Executing Party, the relevant ISO20022 code will be supplied

⁷ 'To Amount' will always be supplied with a 'To Amount Currency' and a 'From Amount' will always be supplied with a 'From Amount Currency', as provided by the Executing Party.

⁸ If the Executing Party supplies a Date Only, and not a Date Time, Calastone will default to 00:00:00 of the date received.

⁹ Use of OTHER, specifically the Proprietary Fund ID Source used, may not be supported by all counterparties. You agree should whether the type of fund identifier (outside of ISIN, SEDOL, APIR or CUSIP) can be supported by each Ordering Party before connecting to the them. If an order is received from an Order Party containing a fund ID designated under OTHER, CTN will send it on to the Executing Party

¹⁰ Must be supplied if field 'transactions/fundIdentifier/type' = "OTHER". Else must not be supplied

Order Confirm Example

The below table provides an example of a posting an Order Confirm and the synchronous response provided by the API.

This is an example of confirming a redemption:

Request	POST /orders/{id}/confirm
Body	<pre>[{ "orderReference": "OrderRef1", "transactionDetails": [{ "transactionType": "Redemption", "dealReference": "DealRef1", "units": 100.0, "cashSettlementDate": "2019-11-19", "grossAmount": { "amount": 1234.56, "currency": "GBP" }, "netAmount": { "amount": 1234.56, "currency": "GBP" }, "settlementAmount": { "amount": 1234.56, "currency": "GBP" }, "fundIdentifier": { "type": "ISIN", "identifier": "AA1231231231" }, "dealingPriceDetails": { "type": "NAVL", "amount": 34.567, "currency": "GBP" } }] }]</pre>

```

"chargeDetails": [
  {
    "type": "BEND",
    "rate": 0.5,
    "amount": 1.23,
    "currency": "GBP"
  },
  {
    "type": "FEND",
    "rate": 1.0,
    "amount": 0.0,
    "currency": "GBP"
  }
],
"commissionDetails": [
  {
    "type": "FEND",
    "rate": 1.12,
    "amount": 2.34,
    "currency": "GBP"
  },
  {
    "type": "CDPL",
    "rate": 1.1,
    "amount": 0.0,
    "currency": "GBP"
  }
],
"taxDetails": [
  {
    "type": "INHT",
    "amount": 1.11,
    "currency": "GBP",
    "exemptionIndicator": true,
    "exemptionReason": "ExemptionReason",
    "extendedExemptionReason": "ExtendedExemptionReason"
  },
  {
    "type": "CTAX",

```

	<pre> "rate": 1.01, "amount": 0.0, "currency": "GBP", "exemptionIndicator": false }], "accountDetails": { "accountId": "Account1", "subAccount": "SubAccount1", "agentCode": "Agent1" }, "tradeDateTime": "2019-11-16T11:54:04" }], "fxDetails": [{ "toAmount": { "amount": 1345.67, "currency": "EUR" }, "fromAmount": { "amount": 1234.56, "currency": "GBP" }, "unitCurrency": "GBP", "quotedCurrency": "GBP", "exchangeRate": 1.25 }] }] </pre>
--	--

Response	200, OK
Body	<pre>[{ "id": "5925c509-50e2-40f4-ae17-dbb6ce213dac", "status": "Confirmed", ... }]</pre>

Below is an example of an order confirmation for a switch:

Request	POST /orders/{id}/confirm
Body	<pre>[{ "orderReference": "OrderRef2", "transactionDetails": [{ "transactionType": "Subscription", "dealReference": "ANewDealRef1", "units": 10.0, "grossAmount": { "amount": 1001.0, "currency": "GBP" }, "netAmount": { "amount": 1000.0, "currency": "GBP" }, "dealingPriceDetails": { "type": 2, "amount": 1.1, "currency": "GBP" }, "settlementAmount": { "amount": 1002.0, "currency": "GBP" }, "fundIdentifier": { "type": "Apir", "identifier": "A11111BB", "name": "Fund1", "supplementaryIdentifier": "SupId1" }, "accountDetails": { "accountId": "Acct1", "subAccount": "SubAcct1", "agentCode": "Agent1" } }] }]</pre>

```

    },
    "tradeDateTime": "2019-08-15T21:32:52",
    "cashSettlementDate": "2019-08-18",
    "legId": "legId123"
  },
  {
    "transactionType": "Redemption",
    "dealReference": "ANewDealRef1",
    "units": 10.0,
    "grossAmount": {
      "amount": 1001.0,
      "currency": "GBP"
    },
    "netAmount": {
      "amount": 1000.0,
      "currency": "GBP"
    },
    "dealingPriceDetails": {
      "type": 2,
      "amount": 1.1,
      "currency": "GBP"
    },
    "settlementAmount": {
      "amount": 1002.0,
      "currency": "GBP"
    },
    "fundIdentifier": {
      "type": "Apir",
      "identifier": "A11111BB",
      "name": "Fund1",
      "supplementaryIdentifier": "SupId1"
    },
    "accountDetails": {
      "accountId": "Acct1",
      "subAccount": "SubAcct1",
      "agentCode": "Agent1"
    },
    "tradeDateTime": "2019-08-15T21:32:52",
    "cashSettlementDate": "2019-08-18",
  }

```

	<pre> "legId": "legid321" },], "fxDetails": [{ "toAmount": { "amount": 1345.67, "currency": "EUR" }, "fromAmount": { "amount": 1234.56, "currency": "GBP" }, "unitCurrency": "GBP", "quotedCurrency": "GBP", "exchangeRate": 1.25 }] }] </pre>
Response	200, OK
Body	<pre> [{ "id": "5925c509-50e2-40f4-ae17-dbb6ce213dac", "status": "Confirmed", ... }] </pre>

Validation Information and Context

Calastone Validation

When an order message is received by Calastone, validation is applied to catch common errors that would prevent it from being processed. There are several levels of validation, described below.

The first check occurs when the message is received by Calastone, where it is checked to ensure that it complies with the agreed JSON schema. If this check is not passed, the message will be marked as invalid by the system and will not be processed or passed on to the relevant trading counterparty.

Once the first check has been passed, the message is then converted into the Calastone internal format and then checked against the Calastone data dictionary. If this check is not passed, a Calastone Reject message will be sent back to the Ordering Party and the message will not be passed on to the counterparty. Where a Calastone Reject is received, a specific rejection reason will be supplied.

There are then a series of additional validation checks done following this, which are designed to catch common errors. These additional checks are outlined in the table below. Similarly, where any of these checks are not passed, a Calastone Reject message will be sent back to the Ordering Party where relevant and a rejection reason will be supplied.

Error Name	Business Attribute	Description	Order Type		
			Subscription Order	Redemption Order	Switch Order
Missing Fund Provider Check	Fund Provider Identifier	An order must clearly identify the intended fund provider. If the fund provider is not identified then the order will fail validation.	☐	☐	☐
Valid Fund Provider Check	Fund Provider Identifier	An order must be for a fund provider that the Ordering Party has been set up to trade with on Calastone. If the Ordering Party has not been set up to trade with the fund provider then the order will fail validation.	☐	☐	☐
Duplicate Order Check	Order Reference	The Order Reference is reported as a duplicate if any other order has been sent by the same Ordering Party with the same Order Reference within the last 7 days.	☐	☐	☐
Missing Order Reference Check	Order Reference	An order must have a unique Order Reference. If this is not provided, the order will fail validation.	☐	☐	☐
Missing Account Check	Account	If no Account is provided, the order will fail validation. Calastone does not check that the Account itself is valid.	☐	☐	☐
Fund ID Check	Fund ID Fund ID Type	An order must have a valid Fund ID. If this is not provided the order will fail validation.	☐	☐	☐
ISIN Check	Fund ID Fund ID Type	If a Fund ID Type of ISIN has been specified the Fund ID must be 12 characters or the order will fail validation.	☐	☐	☐
Units, Ordered Amount, Percentage of Holdings to be Redeemed Check	Units, Ordered Amount, Percentage of Holdings to be Redeemed	Either Units, Ordered Amount or Percentage of Holdings to be Redeemed (for redemption orders only) must be populated. If all Business Attributes are missing then the order will fail validation. If more than one of these Business Attributes are provided then the order will fail validation.	☐	☐	☐
Percentage Rate Check	Percentage of Holdings to be Redeemed	If a Subscription Order has a Percentage of Holdings to be Redeemed value then the order will fail validation.	☐		☐

Group 1 or 2 Units on a Subscription	Group 1 or 2 Identifier	A Subscription Order or the subscription leg of a Switch Order must not have the type of units defined. If the Group 1 or 2 Identifier is populated then the order will fail validation.	☐		☐
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Order Instruction Status Changes

The following table provides further detail of what the Status Code within the Status field corresponds to. Please read this table in conjunction with the Operating Model earlier on in this document.

Status	Status Code	Description
Calastone Accept	Submitted	Indicates that the order has been received and processed by Calastone, has passed validation and will be sent on to the Executing Party.
Calastone Reject	Invalid	Indicates that the order has failed validation and has not been sent onto the Executing Party. A reason will always be provided indicating why the message has failed Calastone's validation.
Business Accept	Accepted	Indicates that the order has been accepted by the Executing Party for further processing.
Business Reject	Rejected	Indicates that the order has not been accepted by the Executing Party for further processing. The Executing Party must provide a reason for the rejection.
Price Confirmation	Confirmed	Indicates that the order has been executed by the Executing Party.

Calastone Accept / Reject

The information relating to the Calastone Accept or Reject status change is contained within the **Validation Details** ("validationDetails") portion of the JSON.

Switch	Subscription/ Redemption	JSON Field Name	Business Attribute Name	Business Attribute Definition	Field Format
M	M	valid	Validation Indicator	Indicates whether the order is valid.	Type: Code Set Format: MaxLength; 5 Content: Valid: "true" Invalid: "false"
M	M	createdUtc	Validation Time	Indicates the time of validation.	Type: Date Time Format: Length; 17 Content: YYYYMMDD HH:MM:SS (Order creation date time)
M	M	orderReference	Order Reference Switch Reference	Unique and unambiguous identifier for an order, as assigned by the Ordering Party.	Type: String Format: MaxLength; 35 Content: Unique Order Reference
O	O	error	Rejection Reason	Additional information to clarify the status or condition of the investment fund order within the processing life cycle.	Type: String Format: MaxLength; 350 Content: Reason as free text

Business Accept / Reject

The information relating to the Business Accept or Reject status change is contained within the **Accept Details** ("acceptDetails") portion of the JSON.

Switch	Subscription / Redemption	JSON Field Name	Business Attribute Name	Business Attribute Definition	Field Format
M	M	accepted	Business Accept = "true" Business Reject = "false"	Indicates that the order has received a Business Accept or Reject message from the Executing Party who has accepted or rejected the order for further processing.	Type: Code Set Format: MaxLength; 5 Content: 'True' or 'False'
M	M	createdUtc	Status Date Time	The date and time the status message was created.	Type: Date Time Format: Length; 17 Content: YYYYMMDD HH:MM:SS
M	M	orderReference	Order Reference Switch Reference	Unique and unambiguous identifier for an order, as assigned by the Ordering Party.	Type: String Format: MaxLength; 35 Content: Unique Order Reference
O	O	dealReference	Deal Reference	The unique reference for the transaction assigned by the Executing Party	Type: Text Format: MaxLength; 35 Content: Unique Deal Reference
O	O	expectedTradeDateTime	Expected Trade Date Time	Expected date or expected date and time at which a price will be applied.	Type: Date Time Format: Length; 17 Content: YYYYMMDD HH:MM:SS
O	O	expectedCashSettlementDate	Expected Cash Settlement Date	Expected date at which the financial instruments will be exchanged against cash.	Type: Date Format: Length; 8 Content: YYYYMMDD
O	O	rejectReason	Rejection Reason	Additional information to clarify the status or condition of the investment fund order within the processing life cycle.	Type: String Format: MaxLength; 350 Content: Reason as free text

Connectivity

The Executing Party will send and receive JSON via a REST Service.

Calastone will host the service and the Executing Party will need to push orders to Calastone and poll at regular intervals for status and prices.

Prior to UAT, the following details will be exchanged between Calastone and the Executing Party):

- UAT REST URL
- Login credentials
- Test Fund Provider Id

In advance of going live, the following information will be exchanged between Calastone and the Executing Party:

- Production / DR REST URL
- Login credentials
- Fund Provider Id (one for each counterparty that the Executing Party wishes to connect with)

Approvals

Approval of the specification is required from the Executing Party prior to the commencement of UAT and going live with Calastone. This approval should be provided via email.

Handling HTTP Responses

Possible Client Request Error Responses

Below are example response bodies for each possible error.

400 Bad Request

```
{
  "errors": {
    "FundProviderIdentifier": [
      "The FundProviderIdentifier field is required."
    ]
  },
  "type": "https://tools.ietf.org/html/rfc7231#section-6.5.1",
  "title": "One or more validation errors occurred.",
  "status": 400,
  "traceId": "...id..."
}
```

401 Unauthorized

No Body Returned

403 Forbidden

No Body Returned

404 Not Found

```
{
  "type": "https://tools.ietf.org/html/rfc7231#section-6.5.4",
  "title": "Not Found",
  "status": 404,
  "traceId": "...id..."
}
```

422 Unprocessable Entity

```
{
  "RuleDuplicateOrderRef": [
    "Duplicate OrderReference"
  ]
}
```

429 Too Many Requests

Where X = calls and Y = seconds - can change depending on client or endpoint. Occurs when client exceeds request quota limits. This limit is set to 10 POST requests per second and 25 GET requests per second.

API calls quota exceeded! maximum admitted X per Ys.

Possible Server Error Responses

These errors are rare but possible. As such, they should be accounted for and handled by the client.

500 Internal Server Error

No Body Returned

502 Bad Gateway

No Body Returned

503 Service Unavailable

No Body Returned

504 Gateway Timeout

No Body Returned

Appendix I

Charge/Commission Details

The following table details the possible values permitted in the **Charge/Commission Type** field in the order instruction and price confirmation, when its inclusion is relevant.

Commission/Charge Type Code	Description
BEND (BackEndLoad)	Fee paid by the investor to the fund or fund management company when redeeming.
BRKF (BrokerageFee)	Fee paid to a broker for services provided.
COMM (Commission)	Amount paid out by the fund to a third party. Please note Calastone will provide this code for all commissions provided by the counterparty, with the exception of CDPL.
CDPL (CommissionDePlacement)	Amount of cash due to an intermediary for selling a product, or services, to a third party.
CDSC (ContingentDeferredSalesCharge)	Deferred sales charge.
CBCH (CorrespondentBankCharge)	In investment funds, the charge of the correspondent bank for transferring money.
DLEV (DilutionLevy)	In investment funds, a charge payable by the investor covering bid-offer spreads and dealing charges for the underlying investments. The dilution levy is paid to the fund for the benefit of other unit holders.
FEND (FrontEndLoad)	Fee paid by the investor to the fund or fund management company when subscribing.
INIT (Initial)	Charge paid at the time of the first subscription.
ADDF (AdditionalFee)	Additional fee not covered by another fee type.
POST (PostageCharge)	Charge paid for the postage.
PREM (Premium)	In investment funds, pre-arranged addition to the trade amount based on the published net asset value.
CHAR (ServiceProvisionFee)	Fee paid by the investor to a distributor/intermediary or other service provider for the provision of financial services.
SHIP (ShippingCharge)	Charge for shipping, including the insurance of securities.
SWIT (Switch)	In investment funds, charge related to a switch transaction.
UCIC (UCITSCommission)	Charges paid by the investor to the fund company for subscription and redemption orders.
REGF (RegulatoryFee)	Fee charged by a regulatory authority, for example, securities and exchange fees.
PENA (Penalty)	Fee charged to the investor for early redemption of the fund.

Price Details

The following table details the possible values permitted in the **Price Type** field in the price confirmation, when its inclusion is relevant.

Price Type Code	Description
BIDE (Bid)	Price is the calculated bid price of a dual-priced fund (offer-bid prices), ie, the selling price of the units for the investor.
OFFR (Offer)	Price is the calculated offer price of a dual-priced investment fund (offer-bid prices), ie, the buying price of the units for an investor.
NAVL (NetAssetValue)	Price is the net asset value per unit that is used either as a transacting price for a single-priced investment fund class, or as a notional price for the calculation of other prices.
CREA (Creation)	Price is the calculated creation of price of a dual-priced investment fund (creation-cancellation prices) and the highest possible buying price of the units before the initial charge. The actual buying or offer price, which includes charges, will be higher.
CANC (Cancellation)	Price is the calculated cancellation price of a dual-priced investment fund (creation-cancellation price), and the lowest possible valuation of units on any one-day. The actual selling or bid price is usually higher.
INTE (Interim)	Price is the average price between the bid and offer prices.
SWNG (Swing)	Price is the calculated net asset value price of a single-priced investment fund. The price is adjusted to take into account the dealing costs due to individual large deals, or due to a significant imbalance in volumes of subscriptions vs redemptions, as an alternative to dilution levy.
MIDD (Mid)	Price is the average price between the bid and offer prices.
RINV (Reinvestment)	Price is the price used when reinvesting units after distribution.
SWIC (Switch)	Price is the price used when transferring units between products.
DDVR (DailyDividendRate)	Price is the net income of a financial instrument, calculated on each dealing day, and divided by all the units outstanding on that day.
ACTU (Actual)	True offer price.

Tax Details

The following table details the possible values permitted in the **Tax Type** field in the price confirmation when its inclusion is relevant.

Tax Type Code	Description
PROV (Provincial)	Tax is a provincial tax.
NATI (NationalTax)	Tax is a national tax.
STAT (StateTax)	Tax is a state tax.
WITH (WithholdingTax)	Tax is on income deducted at source, which the paying agent is legally obliged to deduct from its payments of interest on deposits and other proceeds of a financial instrument.
KAPA (CapitalGainTax)	Tax is on a capital gain (realised and unrealised), ie, the profit that is gained from the sale of a financial instrument.
NKAP (CapitalLossCredit)	Tax recovery is due to a capital loss.
INPO (InterimProfitTax)	Tax is on the sum of all earnings/revenues accrued since the last dividend distribution.
STAM (StampDuty)	Tax is on certain documents and transactions.
WTAX (WealthTax)	Tax is a special tax imposed, applicable only when the value of assets or categories of assets owned by an entity are above a given ceiling defined by the tax authority. Wealth tax is not linked to income.
INHT (InheritanceTax)	Tax that is payable at the time of death of items (money or otherwise), where ownership changes either upon death or within a legally specified number of years before death.
SOSU (SolidaritySurcharge)	Tax that is levied by legal authorities for solidarity purposes.
CTAX (TaxCredit)	Direct reduction of an individual's tax liability.
GIFT (GiftTax)	Tax that is levied on assets given to individuals prior to death of the donor. Gift tax is designed to ensure the integrity of the inheritance tax, preventing the pre-death transfer of wealth.
COAX (ConsumptionTax)	Tax that is levied on goods and services purchased by customers, and is added to the retail price.
EUTR (EUTaxRetention)	Tax withheld at source in the framework of the European Directive on taxation of savings in the form of interest payments.
AKTI (Aktiengewinn I)	Share-related profit. This is a tax specific to the German market. The share-related-profit gives the fund the opportunity to pass on to the institutional investor

AKT2 (Aktiengewinn2)	redeeming fund units certain tax-free income from equity. The calculation method is based on the old Investment Tax Act (Investmentsteuergesetz).
ZWIS (Zwischengewinn)	Zwischengewinn.

Tax Exemption Details

The following table details the possible values permitted in the **Tax Exemption Reason** field in the price confirmation when its inclusion is relevant.

Tax Exemption Code	Description
NONE (None)	Account is not tax exempt.
MASA (MaxiISA)	Maxi Individuals Savings Account (ISA). Account is an ISA with a single ISA manager that may contain cash, stocks and shares, investment funds and life assurance up to the prescribed annual subscription limit (required for UK). The Maxi ISA allows an investor to save all three of the different elements of ISA investments that are allowed, with one savings institution in a single account.
MISA (MiniCashISA)	Mini Cash Individual Savings Account (ISA). Account is an ISA with a single ISA manager that may contain cash, stocks and shares, investment funds and life assurance up to the prescribed annual subscription limit (required for UK). An investor can hold more than one Mini ISA to utilise the overall Mini ISA investment limit, and can invest with more than one savings institution, however only one of the three elements of ISA investment is allowed in a single account.
SISA (MiniStocksAndSharesISA)	Mini Stocks and Shares Individual Savings Account (ISA). Account is an ISA with an ISA manager that may only contain stocks, shares and collective investments up to the prescribed annual subscription limit (required for UK).
IISA (MiniInsuranceISA)	Mini Insurance Individual Savings Account (ISA). Account is an ISA with an ISA manager that may only contain life insurance investments up to the prescribed annual subscription limit (required for UK).
CUYP (CurrentYearPayment)	Account is for current year payments into an investment retirement account (required for US).
PRYP (PriorYearPayment)	Account is for a prior year payment into an investment retirement account (required for US).
ASTR (AssetTransfer)	Account is for an asset transfer into an investment retirement account (required for US).
EMPY (EmployeePriorYear)	Account if for employee- prior year contributions into an investment retirement account (required in US).

EMCY (EmployeeCurrentYear)	Account is for employee - current year contributions into an investment retirement account (required for US).
EPRY (EmployerPriorYear)	Account is for employer- prior year contributions into an investment retirement account (required for US).
ECYE (EmployerCurrentYear)	Account is for employer- current year contributions into an investment retirement account (required for US).
NFPI (NonFundPrototypeIRA)	Account is for non-fund prototype Individual Retirement Account (IRA) into an investment retirement account (required for US).
NFQP (NonFundQualifiedPlan)	Account is for non-fund qualified plan into an investment retirement account (required for US).
DECP (DefinedContributionPlan)	Account is for a defined contribution plan (required for US).
IRAC (IndividualRetirementAccount)	Account is an Individual Retirement Account (IRA) (required for US)
IRAR (IndividualRetirementAccountRollover)	Account is for an Individual Retirement Account (IRA) – Rollover (required for US).
KEOG (KEOGH)	KEOGH. Account is a tax-deferred retirement plan designed to help self-employed workers or individuals who earn self-employed income establish a retirement savings program (required for US).
PFSP (ProfitSharingPlan)	Account is a Profit Sharing Plan (required for US).
401K (FourHundredAndOneK)	401k. Account is an optional retirement plan supported by many companies, in which income is invested before it is taxed (required in US).
SIRA (SelfDirectedIRA)	Self-Directed Individual Retirement Account (IRA) (required for US).
403B (FourHundredAndThreeB)	403(b). Account is tax-deferred investment and savings program for employees of hospitals, educational institutions and other non-profit organisations, to save and invest in their own retirement. The name refers to the relevant section in the US Internal Revenue Code (required for US).
457X (FourHundredFiftySeven)	457. Account is a non-qualified deferred compensation program that is available to state and local governments and not-for-profit employers (required for US).
RIRA (RothIRAFundPrototype)	Roth Individual Retirement Account (IRA) Fund Prototype. Account is a tax-deferred retirement account based on the reverse principle of a traditional IRA. Retirement contributions are taxed upfront, but withdrawals may be made tax-free when a certain age is reached, provided the ROTH IRA has been set up for a certain period of time (required for US).
RIAN (RothIRANonPrototype)	Roth Individual Retirement Account (IRA) Non-Prototype. Account is a tax-deferred retirement account based on the reverse principle of a traditional IRA.

	Retirement contributions are taxed upfront, but withdrawals may be made tax-free when a certain age is reached, provided the ROTH IRA has been set up for a certain period of time (required for US).
RCRF (RothConversionIRAFundPrototype)	Roth Conversion Individual Retirement Account (IRA) Fund Prototype. Account is a retirement account established when a party switches from a regular IRA (required for US).
RCIP (RothConversionIRANonPrototype)	Roth Conversion Individual Retirement Account (IRA) Non-Prototype. Account is a retirement account established when a party switches from a Regular IRA (required for US).
EIFP (EducationIRAFundPrototype)	Education Individual Retirement Account Fund Prototype (also known as an Education Savings Account). Account is a savings plan open to anyone under the age of 18. When the account owner withdraws money to pay for qualified higher education expenses, the withdrawals are generally tax-free (required for US).
EIOP (EducationIRANonPrototype)	Education Individual Retirement Account Non-Prototype (also known as an Education Savings Account). Account is s savings plan open to anyone under the age of 18. When the account owner withdraws money to pay for qualified higher education expenses, the withdrawals are generally tax-free (required in the US).